

Phase 5 — Detector Benchmark

LineTwin · Accenture Innovation Challenge 2026 · Round 2 · Problem Track 4 "DigitalTwin.ai"

Purpose

Score all six classical bottleneck-detection methods against Phase 4's sensitivity-based ground truth — settling, on our own line and with our own measured ground truth, a comparison the published literature can only ever argue about from real MES data with no ground truth to check against. Also build the production diagnostic (`diagnostic/bottleneck.py`): the live momentary Active Period Method, mode decomposition, Bottleneck Walk phrasing, and a significance annotation.

This phase found and fixed one bug in the simulation core (not the diagnostic code), one bug in the significance layer's use of a third-party library, and two of my own test-writing mistakes — all documented below rather than smoothed over, because each is the kind of thing that would otherwise either silently corrupt a result or silently pass a broken test.

Headline result

Ten seeds, all six detectors, scored against the Phase 4 ground truth (`detector_comparison.csv`):

Detector	Top-1 accuracy	Mean MSE	Verdict
Active Period (avg. duration)	100%	0.00179	Best of the six — matches the literature's expectation
Utilization	100%	0.00459	Also fully correct here, moderate MSE
Busy Ratio	100%	0.00464	Fully correct, similar MSE to Utilization
Queue Length	0%	0.00571	Consistently wrong, in a specific, explicable way
Turning Point	100%	N/A (discrete)	Fully correct across all 10 seeds
Arrow	70%	N/A (discrete)	Genuinely unstable — a real finding, not noise

Active Period Method having the lowest MSE among the four continuous-score methods is an honest, independently-measured echo of Roser & Nakano's published ranking (APM 0.04% vs. Utilization 29.21% in their table) — arrived at with a completely different methodology (our own ground truth, our own MSE definition, our own line topology), which is exactly the kind of convergent evidence that is more credible

than either result alone. See `docs/CITATIONS.md` for why we do not claim to have reproduced their exact numbers.

On Queue Length's 0% and Arrow's 70%: both findings survived a serious investigation that changed the simulation core itself (below). They are published as measured, not smoothed into a better-looking number.

Bug found in the simulation core: the unconstrained source

While building the Queue Length detector, it picked **S01 — the very first station — in 100% of runs (5/5 seeds)**, with S01's own input buffer sitting at a measured 5.9975/6.0 capacity: permanently, almost exactly full. That is not a struggling station; it is a saturated one, and investigating why found a real bug: `Line._source()` fed units into the first station **as fast as its buffer would accept them** — an unconstrained, instantaneous upstream supply, not a modelled arrival process.

This is a genuine confound, not a detector-implementation problem: an infinitely-fast source makes the first station look bottleneck-like (near-permanently busy, near-permanently full queue) regardless of whether it is actually resource-constrained. Worse, it turned out to be corrupting **two** of six detectors at once — Queue Length (100% wrong) and, on inspection, the **production momentary Active Period Method itself**: sampled 80 times across a normal run, the live momentary rule split **54% S01 / 45% S17** between the artifact and the true bottleneck.

Fix: arrivals are now paced with the same lognormal sampling used everywhere else in the line, at the first station's own zone (mean, cv) — modelling an upstream supply process with its own natural variability rather than an unconstrained tap. Stamped `synthetic — uncalibrated`, same discipline as every other timing parameter in this project (`scenarios/line30.yaml`).

Verified fixed: S01's mean input-queue occupancy dropped to 3.3–4.8/6 across seeds (no longer pinned at capacity), and the momentary rule's S01 share dropped from 54% to 5% across the same 80-sample protocol. The full test suite (all 70 tests, spanning Phases 1–4) was re-run after this change and stayed green, and Phase 4's ground truth was recomputed from scratch — S17 remains the ground-truth bottleneck, and the shifting-bottleneck trace is qualitatively unchanged (S17 dominant through 60% SUV concentration, S19 takes over at 80%+). The fix corrected an artifact; it did not change the underlying finding.

What the fix revealed instead: two real, explicable detector weaknesses

Queue Length now consistently picks S12 — the last body-zone station, immediately upstream of the bottleneck's paint-zone. This is a well-understood, legitimate weakness of the method, not a new artifact: queue occupancy accumulates progressively through a long, near-balanced section approaching a real constraint (a standard queueing-theory effect — blocking probability climbs moving upstream in a line

with finite buffers, even where no station there is actually rate-limited), so the method detects the *shadow* of the bottleneck one station early rather than the bottleneck itself. Verified stable across seeds — a systematic miss, not noise (`test_queue_length_consistently_misses_in_the_documented_way`).

Arrow ties/misses in a specific, characterizable way. Diagnosing a 2-vs-2 net-in-degree tie between S13 and S17 in one seed traced to the same underlying phenomenon: blocking probability climbs diffusely across the entire near-balanced body zone before sharply reversing at the true bottleneck, and the pairwise Arrow rule accumulates a competing "sink" wherever the first reversal happens to occur, not necessarily at the dominant one. Across 10 seeds this produced 7 correct (S17) and 3 incorrect (S13) picks — genuine, seed-dependent instability rather than a fixed bug, and it is asserted as a *rate* (`test_arrow_is_measurably_unstable_across_seeds`), not forced to pass or fail on one run.

Both are published as measured. Neither was "fixed" by adjusting a tie-break rule to the answer we wanted — that would have been exactly the kind of dishonesty this project's citation ledger exists to prevent.

Bug found in the significance layer: `numpy.bool` is not Python's `bool`

Writing a direct unit test for `significance_annotation()` — ten short, tightly-clustered synthetic active periods for a "bottleneck" station clearly separated from ten similarly-tight periods for two "other" stations (100 vs. ~20, obviously significant) — the function returned "provisional" when it should have returned "established".

Root cause: `statsmodels.stats.multicomp.pairwise_tukeyhsd`'s summary table stores its `reject` column as `numpy.bool`, not Python's built-in `bool`. The code checked `if reject is not True`, an identity comparison that is **always** `True` for a `numpy.bool` value — even when `bool(reject)` is `True` — so `significant_against_all` was being set to `False` on effectively every call, regardless of the actual statistical result. Fixed to `if not bool(reject)`. This is exactly the kind of bug a smoke test would never catch (the live simulation never naturally produces a case clean enough to notice the difference — see below) and only a synthetic, deliberately-separated test fixture surfaces.

A structural, honest limitation found — not a bug, and not fixed

Investigating why `significance_annotation()` returns "provisional" on essentially every sample of the live 30-station line (even once the `numpy.bool` bug was fixed) led to a real, counterintuitive, and permanent characteristic of this approach, not a defect:

A genuine, dominant bottleneck has few, very long active periods, not many, short ones — because by definition it rarely stops being active. S17, sampled at t=20,000s, had accumulated exactly 2 active periods: one 68.3s period early in the run, and one 19,097.2s period covering nearly the entire rest of it. `MIN_PERIODS_FOR_SIGNIFICANCE = 3` per station is a defensible statistical minimum before attempting ANOVA at all — but a station *this* dominant will almost never clear it, precisely because it is dominant.

Weaker, more marginal candidates that flicker between active and inactive more often clear the replication threshold more easily, which is backwards from what a reader would intuitively expect "provisional" to mean.

This was not patched by lowering the threshold to force "established" to appear more often — running ANOVA on $n=1-2$ samples per group would be statistically meaningless, exactly the kind of manufactured confidence this project's honesty discipline exists to prevent. Instead: `test_significance_reaches_established_when_evidence_clearly_supports_it` proves the ANOVA + Tukey-Kramer wiring is correct on data structured to actually test it, decoupled from whether the live simulation happens to produce enough replication naturally. The counterintuitive behavior itself is documented in `bottleneck.py`'s docstring and here, so a future reader encountering "provisional" on an obvious bottleneck understands why, rather than assuming the significance layer is broken.

Two of my own test-writing mistakes, also documented rather than quietly fixed

- `bottleneck.py`'s own docstring illustrated mode decomposition with `{"working": 0.59, "repair": 0.41}`, captioned "reads as downtime-dominant." It does not — $0.59 > 0.41$, so `working` is the dominant mode by the function's own (correct) logic. The docstring's example was wrong from the start; fixed to `{"working": 0.41, "repair": 0.59}`, and the corresponding test corrected to match.
 - The live momentary rule genuinely contests close to 50/50 between S17 and **S13** (the paint zone's higher-CV gateway station) at arbitrary sampling instants — not a bug, but a real property of a live, instantaneous signal, and exactly why the batch "average active duration" method (100% correct, 5/5 and 10/10 seeds) exists for retrospective comparison.
`test_momentary_rule_is_dominated_by_the_two_genuine_contenders` asserts this contested-but-bounded reality ($S17+S13 > 85\%$ of picks) rather than asserting a single "always-correct" answer the live rule does not actually provide.
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Deliverables produced

Artefact	What it fixes
<code>src/twin/diagnostic/run_stats.py</code>	Shared statistics substrate (<code>LineRunStats</code>) every detector reads from, so the comparison is fair
<code>src/twin/diagnostic/detectors.py</code>	Six detectors behind one interface: Utilization, Active Period, Busy Ratio, Queue Length (continuous scores), Arrow, Turning Point (discrete picks)
<code>src/twin/diagnostic/evaluate.py</code>	MSE / top-1 / top-3 scoring against ground truth, with MSE/top-3 correctly <code>None</code> (not invented) for the two discrete-pick methods

Artefact	What it fixes
<code>src/twin/diagnostic/bottleneck.py</code>	Production diagnostic: momentary APM, mode decomposition, Bottleneck Walk phrasing, ANOVA+Tukey-Kramer significance annotation
<code>src/twin/sim/line.py</code> (fix)	Realistic paced arrivals, replacing the unconstrained source bug described above
<code>src/twin/sim/station.py</code> (addition)	Public <code>active_period_start</code> property, replacing a private-attribute reach-around
<code>tools/run_detector_benchmark.py</code>	Produces <code>detector_comparison.csv</code>
<code>docs/phases/detector_comparison.csv</code>	The committed 10-seed × 6-detector result
<code>tests/test_bottleneck.py</code> , <code>tests/test_detectors.py</code>	26 new tests: hand-constructed fixtures for exact logic, synthetic data for the statistical wiring, and the live 30-station line for emergent behavior
<code>pyproject.toml</code>	<code>statsmodels>=0.14</code> added explicitly (verified absent transitively) for Tukey-Kramer

Exit criteria

Criterion	Status
All six detectors implemented and scored against Phase 4's ground truth	Met — <code>detector_comparison.csv</code> , 10 seeds
Result published whatever it shows	Met — Queue Length's 0% and Arrow's 70% are both reported plainly, with cause
Significance layer implemented and its statistical correctness proven	Met — <code>numpy.bool</code> bug found and fixed via a direct synthetic test; the "provisional-on-obvious-bottlenecks" behavior documented as structural, not patched
Mode decomposition + Bottleneck Walk phrasing	Met
Detector test parametrised across stations	Met — 6 clear non-contenders in <code>test_bottleneck.py</code> , plus the full 30-station live line in <code>test_detectors.py</code>
Full suite green after the source-model fix	Met — 70 tests total (44 from Phases 1–4 + 26 new this phase), all green

70/70 tests passing.

(A correction, in the same spirit as the rest of this record: a draft of this document originally claimed "96/96," from "70 prior + 26 new" — arithmetic that double-counted, since the 70 being referenced

already included this phase's 26 new tests. Verified precisely with `git stash -u` to isolate the true pre-Phase-5 state (44 tests) before writing the corrected figure here.)

Next

Per instruction, the build stops here. **Phase 6 — Engine & Transport** is next when resumed: the real-time-paced asyncio tick loop, the single-slot SSE conflation bus, and the REST control surface — building on `diagnostic/bottleneck.py`'s `diagnose()` as the live twin's actual bottleneck feed.